

## **rSumSqDigits**

Write a **recursive** C function that takes a positive integer argument and returns the sums of the squares of the digits of the integer. For example, given the number 3418, the function should return 90, i.e.  $3*3+4*4+1*1+8*8$ . The function prototype is given as follows:

```
int rSumSqDigits(int num);
```

A sample program template is given below for testing the function:

```
#include <stdio.h>
int rSumSqDigits(int num);
int main()
{
    int number;

    printf("Enter a number: \n");
    scanf("%d", &number);
    printf("rSumSqDigits(): %d\n", rSumSqDigits(number));
    return 0;
}
int rSumSqDigits(int num)
{
    /* Write your code here */
}
```

Some sample input and output sessions are given below:

- (1) Test Case 1  
Enter a number:  
123  
rSumSqDigits(): 14
- (2) Test Case 2  
Enter a number:  
3418  
rSumSqDigits(): 90
- (3) Test Case 3  
Enter a number:  
13579  
rSumSqDigits(): 165